1. Introduction

This project presents an advanced WAN network scenario built in Cisco Packet Tracer, designed to compare and analyze three different routing protocols operating within the same environment. The simulation replicates how a real-world WAN network functions, where different segments of the infrastructure leverage different protocols — closely mirroring the architecture used by real ISPs and enterprise networks.

The primary objectives of this activity are:

- Understand and compare the behavior of EIGRP, OSPF, and BGP in a live simulation
- Implement and verify route redistribution between heterogeneous routing domains
- Configure NAT/PAT for private LAN segments connecting to a shared WAN
- Apply systematic troubleshooting methodologies to identify and resolve routing issues

> *NOTE: This lab was designed to reflect real-world network complexity. Each protocol group operates in isolation and only exchanges reachability information through explicitly configured route redistribution at boundary routers.*

2. Network Topology

2.1 Overall Structure

The network is composed of three main router groups, each running a dedicated routing protocol. They are interconnected via four transit routers, and two edge routers handle NAT at the LAN boundaries.

Group	Routers	Protocol	Topology
Left Group	R_L1, R_L2, R_L3, R_L4, R_L5	EIGRP	Full Mesh
Center Group	R_C1 through R_C6	OSPF	Partial Mesh
Right Group	R_R1, R_R2, R_R3, R_R4, R_R5	BGP (eBGP)	Full Mesh
Transit Routers	R_INTER1, R_INTER2, R_INTER3, R_INTER4	Mixed	Point-to-Point
Edge Routers	RouterL, RouterR	NAT/PAT	LAN Gateway

R_INTER1 and R_INTER2 connect the EIGRP and OSPF domains, using GigabitEthernet toward the EIGRP side and Serial links toward OSPF. R_INTER3 and R_INTER4 bridge the OSPF and BGP domains using the same interface convention. Each edge router (RouterL and RouterR) is connected to a managed switch with four client PCs organized into four VLANs.

2.2 IP Addressing Plan

The addressing scheme is designed for readability: each address block immediately reveals which routing domain it belongs to. All point-to-point links use /30 masks to minimize address waste.

Address Block	Mask	Usage
10.1.x.x	/30	Internal links — EIGRP group
10.20.x.x	/30	Internal links — OSPF group
10.30.x.x	/30	Internal links — BGP group
10.12.x.x	/30	GigabitEthernet links to transit routers
10.100.x.x	/30	Serial links on transit routers
192.168.x.x	/24	LAN segment — RouterL (Left side)
172.16.x.x	/24	LAN segment — RouterR (Right side)

3. Routing Protocols

3.1 EIGRP — Enhanced Interior Gateway Routing Protocol

EIGRP was selected for the left router group. It is a Cisco-proprietary protocol that combines characteristics of both distance-vector and link-state protocols, commonly classified as a hybrid protocol. EIGRP uses the DUAL (Diffusing Update Algorithm) to compute optimal routes and maintain pre-calculated backup paths (feasible successors), enabling sub-second failover.

Configuration Overview

All five routers in the EIGRP group run process ID 1. Networks are advertised using the

`network` command with wildcard masks. The `no auto-summary` command is critical to prevent automatic route summarization, which would cause reachability issues in VLSM environments.

Verification — show ip eigrp neighbors (R_L1)

> *R_L1 established EIGRP neighbor adjacencies with three routers across three distinct interfaces. SRTT (Smooth Round-Trip Time) = 40 ms indicates healthy path latency. RTO (Retransmission Timeout) = 1000 ms is the window before packet retransmission. Q Cnt = 0 confirms no queued updates — the network has fully converged via DUAL.*

Key Characteristics

- Fastest convergence among the three protocols tested
- Uses bandwidth and delay as default composite metrics
- Supports unequal-cost load balancing via the variance command
- Cisco-proprietary: limited to homogeneous Cisco environments

3.2 OSPF — Open Shortest Path First

OSPF is a link-state protocol deployed in the center group. Unlike EIGRP, it is an open standard (RFC 2328) and is widely adopted across multi-vendor environments. OSPF builds a complete topology map via Link State Advertisements (LSAs) and computes shortest paths using Dijkstra's algorithm (SPF).

Configuration Overview

The lab uses a single-area design (Area 0 — backbone area), which is appropriate for the scale of the simulation. All networks are advertised with the

`network` command. OSPF automatically elects a **DR (Designated Router)** and **BDR (Backup Designated Router)** on each multi-access segment to reduce LSA flooding overhead.

Verification — show ip ospf neighbor (R_C6)

> *All three OSPF neighbors are in FULL state — the expected final state confirming that LSA exchange is complete and all routers share an identical LSDB. Dead Time resets on every received Hello packet; a neighbor is declared down when this timer reaches zero.*

Key Characteristics

- Open standard — vendor-agnostic and widely supported
- Hierarchical design with multiple areas supported
- Automatic DR/BDR election reduces flooding on broadcast segments
- Slower convergence than EIGRP on large topologies due to SPF recalculation

3.3 BGP — Border Gateway Protocol

BGP is the protocol that holds the internet together. It is a path-vector protocol designed to exchange routing information between Autonomous Systems (AS). In the right group, eBGP (external BGP) is simulated — each router is assigned a unique AS number (AS 301 through AS 305).

Configuration Overview and CPT Limitation

> *IMPORTANT: Cisco Packet Tracer does not support iBGP (all routers sharing the same AS number). Attempting to configure iBGP produces the error: 'BGP is already running; AS is X'. The workaround was to assign each router a unique AS number, which more closely models a real-world multi-ISP topology.*

BGP requires manually specifying neighbors using the

`neighbor` command and their respective AS numbers. A TCP session on port 179 is established before any route exchange begins (three-way handshake). BGP does not advertise networks automatically — all prefixes must be explicitly defined with the `network` command or via redistribution.

Verification — show ip bgp summary (R_R2)

> *R_R2 maintains five active eBGP sessions: three toward routers within the BGP group and two toward transit routers (AS 901 and AS 902). The State/PfxRcd column shows 4 prefixes received from each peer, consistent with the number of networks advertised via the network command on each router. Up/Down timers confirm session stability.*

Key Characteristics

- Only protocol designed for inter-AS routing at internet scale
- Path-vector: uses AS-PATH attribute to prevent routing loops
- Slow convergence relative to EIGRP and OSPF
- Highly flexible policy control via route maps and attributes
- Establishes sessions over TCP/179 — requires full layer-3 reachability first

3.4 Protocol Comparison

Feature	EIGRP	OSPF	BGP
Type	Hybrid (Distance-Vector / Link-State)	Link-State	Path-Vector
Standard	Cisco Proprietary	Open (RFC 2328)	Open (RFC 4271)
Algorithm	DUAL	Dijkstra (SPF)	Best-Path Selection
Convergence	Very Fast	Fast	Slow
Transport	IP (88)	IP (89)	TCP (179)
Use Case	Enterprise (Cisco)	Enterprise / ISP IGP	Inter-AS / Internet
AD (Internal)	90	110	20 (eBGP)
AD (External)	170	110 (E2: 110)	200 (iBGP)

4. Route Redistribution

Since three heterogeneous routing protocols operate within the same topology, route redistribution is required for reachability across protocol boundaries. Redistribution allows a router running two protocols simultaneously to import routes learned by one and advertise them into the other.

4.1 EIGRP ↔ OSPF (R_INTER1 & R_INTER2)

R_INTER1 and R_INTER2 each have one interface in the EIGRP domain (GigabitEthernet) and one in the OSPF domain (Serial). Redistribution is configured bidirectionally.

Issue Encountered — Infinite Metric

> *During initial configuration, EIGRP routes redistributed into OSPF received a metric value of 4,294,967,295 (0xFFFFFFFF — maximum 32-bit value), rendering them unreachable. Root cause: no explicit metric was provided in the redistribute command. Fix: appended 'metric 20 subnets' to the redistribute eigrp 1 command within the OSPF process.*

The

`subnets` keyword is also mandatory: without it, OSPF only redistributes classful networks, silently dropping all /30 subnets used in this topology.

Verification — show ip route (R_C3)

> *R_C3, positioned at the center of the topology, displays route types O (OSPF internal), O E2 (OSPF external type 2 — redistributed from EIGRP on the left and from BGP on the right). The same external prefixes appear via two next-hop addresses, confirming OSPF automatic equal-cost load balancing across R_INTER1 and R_INTER2.*

4.2 OSPF ↔ BGP (R_INTER3 & R_INTER4)

R_INTER3 (AS 901) and R_INTER4 (AS 902) run both OSPF and BGP simultaneously. Redistribution is configured in both directions:

- OSPF → BGP: redistribute ospf 1 metric 20 inside the BGP process
- BGP → OSPF: redistribute bgp 901/902 subnets inside the OSPF process

Issue Encountered — BGP Prefixes Not Propagating

> *BGP routes (10.30.x.x) were absent from OSPF and EIGRP routing tables. Root cause: routers R_R1 through R_R5 were missing the 'network' command in their BGP configuration. BGP does not auto-discover networks from the routing table — every prefix must be explicitly defined. Once network statements were added to all five BGP routers, routes propagated correctly through the full topology.*

End-to-End Verification — show ip route (R_L1)

> *R_L1 (EIGRP-only router) displays routes to 10.20.x.x (OSPF domain), 10.30.x.x (BGP domain), and 172.16.x.x (remote LAN) — all learned as D EX (EIGRP External, AD 170). This confirms that multi-hop redistribution across all three protocols is functioning correctly. Internal EIGRP routes retain AD 90 and are always preferred over external equivalents.*

4.3 Administrative Distance Summary

Route Type	Administrative Distance	Preference
Connected	0	Highest (always preferred)
Static	1	Very High
EIGRP Internal (D)	90	High
OSPF (O)	110	Medium-High
EIGRP External (D EX)	170	Low
OSPF External (O E2)	110	Medium-High
eBGP	20	Very High (between AS)

5. Network Address Translation (NAT)

5.1 Configuration

NAT was configured on RouterL and RouterR — the edge routers connecting private LAN segments to the WAN. PAT (Port Address Translation), also known as NAT Overload, is used. All LAN clients share a single public IP address (the router's WAN interface IP) and are differentiated by unique port mappings.

The configuration consists of three components:

- Mark LAN-facing interfaces as ip nat inside
- Mark the WAN-facing interface as ip nat outside
- Define an ACL identifying the private address ranges to be translated, then bind it with: ip nat inside source list 1 interface GigabitEthernet0/1 overload

Sub-Interface Complication (RouterL)

> *RouterL uses sub-interfaces for inter-VLAN routing (G0/0.10, G0/0.20, G0/0.30, G0/0.40). Initial NAT configuration applied 'ip nat inside' only to the parent interface G0/0, which does not carry actual traffic — traffic flows through sub-interfaces. Solution: 'ip nat inside' was explicitly added to each sub-interface.*

5.2 NAT Translation Table

The

`show ip nat translations` command on RouterL displays active translation entries, confirming PAT is operational. Each entry maps an inside local address (private IP + port) to an inside global address (WAN IP + unique port), maintaining state for return traffic.

5.3 Security and Functional Impact

NAT has a direct impact on network behavior and security posture:

- Private IP addresses (192.168.x.x, 172.16.x.x) are never exposed on the WAN — all traffic appears to originate from the router's single public IP
- Internal network topology is hidden from external observers
- Inbound connections to NAT clients are blocked by default — an external host cannot initiate a session to a client without a static NAT mapping or port forwarding rule
- NAT violates the IP end-to-end principle and can cause issues with protocols that embed IP addresses in the payload (e.g., FTP, SIP/VoIP, some P2P applications)
- NAT is not a substitute for a firewall — it provides obscurity, not authentication or deep packet inspection

> *IPv6 Perspective: NAT was introduced primarily as a workaround for IPv4 address exhaustion. In IPv6, the address space is large enough that NAT is theoretically unnecessary. However, NAT remains prevalent in practice even in hybrid IPv4/IPv6 deployments.*

6. Transport Layer in the WAN Context

6.1 TCP Usage

TCP (Transmission Control Protocol) is used for all control-plane traffic requiring reliable, ordered delivery:

- BGP neighbor sessions run over TCP/179 — a full three-way handshake must complete before any routing updates are exchanged
- BGP messages (OPEN, UPDATE, KEEPALIVE, NOTIFICATION) are delivered reliably with guaranteed ordering
- NAT translation table management for TCP sessions includes tracking SYN, established, and FIN states

6.2 IP-Native Protocol Transport

OSPF and EIGRP bypass TCP and UDP entirely, operating directly over IP:

- OSPF uses IP protocol number 89 — Hello, DBD, LSR, LSU, and LSAck packets are sent directly over IP, with OSPF implementing its own reliability via explicit acknowledgment
- EIGRP uses IP protocol number 88 — RTP (Reliable Transport Protocol) provides selective reliability only for packets requiring acknowledgment (Updates, Queries, Replies)

6.3 UDP Use Cases

UDP (User Datagram Protocol) is used by supporting network services in this topology:

- DHCP — if configured, client discovery uses UDP broadcast on ports 67 (server) and 68 (client). Loss of a single DHCP Discover is non-critical since clients automatically retransmit
- TFTP — used for IOS image transfers during configuration backup (port 69)
- SNMP — network monitoring traffic (port 161/162)

7. Troubleshooting Log

7.1 BGP iBGP Not Supported in Cisco Packet Tracer

Symptom	Error: 'BGP is already running; AS is X' when attempting to add a neighbor in the same AS
Root Cause	CPT does not support iBGP (multiple routers sharing one AS number)
Resolution	Assigned unique AS numbers (301–305) to each router, simulating an eBGP multi-AS topology

7.2 EIGRP Routes Redistributed with Infinite Metric into OSPF

Symptom	EIGRP routes appear in OSPF table with metric 4,294,967,295 — effectively unreachable
Diagnosis	show ip route on OSPF routers showed O E2 routes with maximum metric value
Root Cause	Missing explicit metric in the redistribute eigrp 1 command — OSPF cannot derive a metric from EIGRP's composite metric
Resolution	Added 'metric 20 subnets' to the redistribute command; subnets keyword ensures /30 routes are included

7.3 BGP Prefixes Not Propagating to OSPF and EIGRP Domains

Symptom	10.30.x.x routes absent from OSPF and EIGRP routing tables even after redistribution was configured
Diagnosis	show ip bgp on transit routers showed empty BGP tables — no routes received from the BGP group
Root Cause	Routers R_R1 through R_R5 were missing the 'network' command — BGP does not auto-advertise connected networks
Resolution	Added explicit network statements to all five BGP routers. Routes propagated immediately through the full topology

7.4 NAT Not Applied on Sub-Interfaces

Symptom	VLAN clients on RouterL could not reach WAN destinations despite correct routing
Diagnosis	show ip nat translations showed no active entries for client IPs
Root Cause	'ip nat inside' was configured on G0/0 (parent interface) instead of on sub-interfaces G0/0.10–G0/0.40
Resolution	Added 'ip nat inside' to each VLAN sub-interface individually

8. Conclusions and Key Takeaways

This lab demonstrated in a practical and measurable way how WAN routing works in a heterogeneous multi-protocol environment. Key lessons learned:

- Route redistribution is non-trivial. Metric translation between protocols requires explicit values; silent failures (like the infinite-metric OSPF issue) demand careful verification at every protocol boundary.
- BGP behaves fundamentally differently from IGPs. It is not designed for fast convergence — it is designed for policy-driven, scalable, loop-free inter-domain routing. Understanding its TCP-based session model and explicit network advertisement requirement is essential.
- NAT on sub-interfaces requires per-sub-interface configuration. The NAT inside/outside model does not automatically cascade from parent to child interfaces in IOS.
- Systematic troubleshooting is the only reliable approach. Working from source to destination, router by router, using show ip route and show ip bgp at each hop, is far more effective than end-to-end ping attempts.
- Protocol selection depends on the environment: EIGRP for homogeneous Cisco deployments requiring fast convergence; OSPF for open, scalable enterprise networks; BGP exclusively for inter-AS and internet-facing routing.

> *The topology built in this lab mirrors a simplified but architecturally realistic ISP/enterprise boundary model: internal IGPs (EIGRP and OSPF) handling intra-domain routing, and BGP managing inter-domain reachability — with NAT providing address translation at the customer edge.*